IGCSE 0607 Practice Questions — E10.5 — Combined Questions & Markscheme

Averages on a calculator

Total Questions: 15 — **Total Marks:** 90 Marks

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1 The number of text messages received by 15 students in one hour is shown below.

12	23	15	18	21	14	17	15	22	19	15	24	11	16	20

(a) Find the mean.

..... **[1]**

(b) Find the median.

..... **[1]**

(c) Find the interquartile range.

..... **[2]**

2 A survey was conducted to find the number of books read by 21 students during a month.
The results are shown in the table.

Number of books	1	2	3	4	5
Frequency	2	5	7	4	3

(a) Use a graphic display calculator to find the following statistics for this data.

(i) Calculate the mean number of books read.
Give your answer correct to 3 significant figures.

Mean = ... **[2]**

(ii) Find the median.

Median = ... **[1]**

(iii) Find the interquartile range.

Interquartile range = ... **[2]**

3 The marks scored by 15 students in a mathematics test are shown below.

15	19	14	28	11	16	9	20	15	13	17	21	10	18	14

(a) Find the mean.

..... **[2]**

(b) Find the median.

..... **[1]**

(c) Find the interquartile range.

..... **[2]**

4 A student records the number of hours he spends playing sports each week for 14 weeks.
The results are:

5 3 8 4 5 6 7 3 4 7 5 2 6 5

(a) Find the mode.

..... **[1]**

(b) Find the mean.

..... **[1]**

(c) Find the median.

..... **[1]**

(d) Find the lower quartile.

..... **[1]**

(e) Find the upper quartile.

..... **[1]**

5 The times, in minutes, taken by 15 students to complete a logic puzzle are listed below.

28 17 35 22 46 14 31 25
18 33 26 20 38 23 29

(a) Use your graphic display calculator to find

(i) the mean,

..... minutes **[1]**

(ii) the median,

..... minutes **[1]**

(iii) the lower quartile and the upper quartile.

Lower quartile = ... , Upper quartile = ... **[2]**

(b) Write down the interquartile range.

..... minutes **[1]**

(c) Another student also completes the puzzle in x minutes.
When this time is included, the mean time for all 16 students becomes 27.5 minutes.
Calculate the value of x.

$x =$ ..... **[2]**

6 The table shows the heights, h cm, of 120 plants in a greenhouse.

Height (h cm)	$130 < h \leq 140$	$140 < h \leq 150$	$150 < h \leq 160$	$160 < h \leq 170$	$170 < h \leq 180$
Frequency	15	38	42	18	7

(a) Write down the modal class.

..... **[1]**

(b) Use your graphic display calculator to find an estimate of the mean height.

..... cm **[2]**

7 The table shows information about the time, t minutes, taken by 100 people to travel to work.

Time (t minutes)	Frequency
$0 < t \leq 10$	14
$10 < t \leq 20$	28
$20 < t \leq 30$	35
$30 < t \leq 40$	18
$40 < t \leq 60$	5

(a) Write down the modal class.

..... **[1]**

(b) Use your graphic display calculator to calculate an estimate of the mean time.

..... minutes **[3]**

(c) Explain why your answer to part (b) is an estimate.

..... **[1]**

8 The table shows the time, t minutes, taken by 80 people to complete a crossword puzzle.

Time (t minutes)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$	$40 < t \leq 60$
Frequency	12	25	28	11	4

(a) Calculate an estimate of the mean time.

..... minutes **[3]**

(b) Explain why your answer to part (a) is an estimate.

..... **[1]**

(c) An 81st person completes the puzzle in 12 minutes.
Without calculating the new mean, explain how this affects the mean time.

..... **[1]**

9 A survey was conducted on the time, t minutes, taken by 80 students to travel to school. The results are shown in the frequency table.

Time (t minutes)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$	$40 < t \leq 50$
Frequency	12	25	28	11	4

(a) Calculate an estimate of the mean time.

..... minutes **[2]**

(b) Explain why your answer to part (a) is an estimate.

.....

..... **[1]**

(c) The mean time for a different group of 120 students to travel to school is 24.5 minutes. Calculate the overall mean time for the 200 students.

..... minutes **[3]**

10 A delivery company records the number of parcels, n, delivered by each of 60 vans on Monday. The results are shown in the table below.

Parcels (n)	18	19	20	21	22	23	24
Frequency	4	9	14	15	10	5	3

(a) Use your graphic display calculator to find

(i) the mean number of parcels delivered per van,

..... **[1]**

(ii) the median,

..... **[1]**

(iii) the interquartile range.

..... **[2]**

(b) The company also records the masses, m kilograms, of a sample of 50 parcels. The grouped frequency table shows the results.

Mass (m kg)	Frequency
$0 < m \leq 2$	8
$2 < m \leq 5$	17
$5 < m \leq 10$	15
$10 < m \leq 20$	10

(i) Use your graphic display calculator to calculate an estimate of the mean mass of these 50 parcels.

..... kg **[3]**

(ii) Write down the modal class.

..... **[1]**

11 The table shows the number of goals scored by a football team in 35 matches.

Number of goals	0	1	2	3	4	5
Frequency	3	x	y	6	4	1

(a) The mean number of goals scored is 2.2.
Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(b) Find the median number of goals.

..... **[1]**

12 The table shows information about the number of goals scored by a football team in a number of matches.

Number of goals	0	1	2	3	4
Frequency	6	x	8	4	2

The mean number of goals scored is 1.5.

(a) Show that $x = 12$.

[3]

(b) Find the median.

..... **[1]**

(c) Find the interquartile range.

..... **[2]**

13 A game is played where players can score $1, 2, 3, 4$, or 5 points.
The table shows the frequencies of the scores obtained by a group of players.
The mean score is 2.8.

Score	1	2	3	4	5
Frequency	10	k	$2k+3$	7	5

(a) Show algebraically that $k = 5$.

[4]

(b) Use your graphic display calculator to find the following statistics for the scores.

(i) Find the median.

Median = ... **[1]**

(ii) Find the lower quartile.

Lower quartile = ... **[1]**

(iii) Find the upper quartile.

Upper quartile = ... **[1]**

14 The number of days of rain per week in a city was recorded for 40 weeks.
The table shows the results.

Number of days	0	1	2	3	4
Frequency	8	x	y	6	2

The mean number of days of rain per week is 1.55.

(a) Show that $x + y = 24$ and $x + 2y = 36$.

[3]

(b) Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[2]**

(c) Use your graphic display calculator to find:

(i) the median

..... **[1]**

(ii) the interquartile range

..... **[1]**

15 A weather station records the daily rainfall, r millimetres, over 100 days.
The grouped frequency table shows the results, where p and q are positive integers.

Rainfall (r mm)	Frequency
$0 \leq r < 5$	34
$5 \leq r < 10$	p
$10 \leq r < 20$	26
$20 \leq r < 30$	q
$30 \leq r < 50$	8

(a) Use the total number of days to show that $p + q = 32$.

[1]

(b) Using the mid-interval values, the estimated mean daily rainfall for the 100 days is 12.45 mm.

(i) Show that $3p + 10q = 180$.

[3]

(ii) Solve the simultaneous equations $p + q = 32$ and $3p + 10q = 180$ to find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[2]**

(c) For the 12 days in the interval $20 \leq r < 30$, the exact daily rainfall values, in millimetres, are listed below.

24 27 21 28 22 25 20 27 23 28 22 24

Use your graphic display calculator to find, for these 12 days,

(i) the mean,

..... mm **[1]**

(ii) the median,

..... mm **[1]**

(iii) the interquartile range.

..... mm **[2]**

(d) When the exact values from part (c) are used instead of the mid-interval value for the interval $20 \leq r < 30$, calculate the revised estimate of the mean daily rainfall for all 100 days.

..... mm **[2]**

1 **(a)** 17.5 (or $17\frac{7}{15}$ or 17.46...) **[B1]**

(b) 17 **[B1]**

(c) Upper quartile = 21 and Lower quartile = 15 **[M1]**

6 **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** **(i)** 3.0476... **[M1]**

3.05 **[A1]**

(ii) 3 **[B1]**

(iii) Upper quartile = 4 and Lower quartile = 2 identified **[M1]**

2 **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** 16 **[B2]**

If B0, M1 for $\sum x = 240$ seen

(b) 15 **[B1]**

(c) 6 **[B2]**

If B0, M1 for $Q_1 = 13$ and $Q_3 = 19$ seen

[Back to Question] *[Total Marks: 5]*

4 **(a)** 5 **[B1]**

(b) 5 **[B1]**

(c) 5 **[B1]**

(d) 4 **[B1]**

(e) 6 **[B1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** **(i)** 27 **[B1]**

(ii) 26 **[B1]**

(iii) Lower quartile = 20 **[B1]**

Upper quartile = 33 **[B1]**

(b) 13 **[B1]**

(c) $(16 \times 27.5) - (15 \times 27) = 440 - 405$ **[M1]**

35 **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** $150 < h \leq 160$ **[B1]**

(b) Uses midpoints $135, 145, 155, 165, 175$ **[M1]**

152 **[A1]**

[Back to Question] *[Total Marks: 3]*

7 **(a)** $20 < t \leq 30$ **[B1]**

(b) Use of correct midpoints: $5, 15, 25, 35, 50$ **[M1]**

$\Sigma fx = 14(5) + 28(15) + 35(25) + 18(35) + 5(50)$ or 2245 seen **[M1]**

22.45 **[A1]**

(c) The exact times are not known so the midpoints of the intervals are used. **[B1]**

[Back to Question] *[Total Marks: 5]*

8 **(a)** Use of correct mid-interval values $(5, 15, 25, 35, 50)$ **[M1]**

$\Sigma ft = 5(12) + 15(25) + 25(28) + 35(11) + 50(4)$ or 1720 **[M1]**

21.5 **[A1]**

(b) The exact times are not known / the data is grouped so mid-interval values are used. **[B1]**

(c) The mean will decrease because 12 is less than the current mean (21.5). **[B1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** 21.25 **[B2]**

If B0, M1 for $\sum fx = 1700$ or midpoints $5, 15, 25, 35, 45$ seen

(b) The exact times are not known / the midpoints of each class are used as an estimate. **[B1]**

(c) $120 \times 24.5 = 2940$ **[M1]**

$(1700 + 2940) \div 200$ **[M1]**

23.2 **[A1]**

[Back to Question] *[Total Marks: 6]*

10 **(a)** **(i)** 20.75 **[B1]**

(ii) 21 **[B1]**

(iii) Upper quartile = 22 and lower quartile = 20 **[M1]**

2 **[A1]**

(b) **(i)** Mid-interval values: $1, 3.5, 7.5, 15$ **[B1]**

$\dfrac{8(1)+17(3.5)+15(7.5)+10(15)}{50} = \dfrac{330}{50}$ **[M1]**

6.6 **[A1]**

(ii) $2 < m \leq 5$ **[B1]**

[Back to Question] *[Total Marks: 8]*

11 **(a)** $3+x+y+6+4+1=35$ leading to $x+y=21$ **[M1]**

Expression for total goals: $0(3)+1(x)+2(y)+3(6)+4(4)+5(1)$ or $x+2y+39$ **[M1]**

$\dfrac{x+2y+39}{35} = 2.2$ leading to $x+2y=38$ **[M1]**

$x=4$ and $y=17$ **[A1]**

(b) 2 **[B1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** $\dfrac{0\times 6+1\times x+2\times 8+3\times 4+4\times 2}{6+x+8+4+2} = 1.5$ **[M1]**

$\dfrac{x+36}{x+20} = 1.5$ **[M1]**

$x+36=1.5x+30 \implies 0.5x=6 \implies x=12$ **[A1]**

(b) 1 **[B1]**

(c) 1 **[B2]**

If B0, M1 for $Q_1=1$ and $Q_3=2$ seen

[Back to Question] *[Total Marks: 6]*

13 **(a)** Total frequency $= 10+k+2k+3+7+5 = 3k+25$ **[M1]**

Total score $= 10(1)+k(2)+(2k+3)(3)+7(4)+5(5) = 8k+72$ **[M1]**

$\dfrac{8k+72}{3k+25} = 2.8$ **[M1]**

$8k+72 = 8.4k+70 \implies 0.4k=2 \implies k=5$ **[A1]**

(b) **(i)** 3 **[B1]**

(ii) 1.5 **[B1]**

(iii) 4 **[B1]**

[Back to Question] *[Total Marks: 7]*

14 **(a)** $8+x+y+6+2=40 \implies x+y=24$ **[B1]**

$\dfrac{0\times 8+1\times x+2\times y+3\times 6+4\times 2}{40} = 1.55$ **[M1]**

$x + 2y + 26 = 62 \implies x + 2y = 36$ **[A1]**

(b) Subtracts equations: $(x + 2y) - (x + y) = 36 - 24 \implies y = 12$ **[M1]**

$x = 12, y = 12$ **[A1]**

(c) **(i)** 1.5 **[B1]**

(ii) 1 **[B1]**

[Back to Question] *[Total Marks: 7]*

15 **(a)** $34 + p + 26 + q + 8 = 100$ simplifying to $p + q = 32$ **[B1]**

(b) **(i)** $\sum fm = 34(2.5) + 7.5p + 26(15) + 25q + 8(40) = 7.5p + 25q + 795$ **[M1]**

$\frac{7.5p + 25q + 795}{100} = 12.45$ giving $7.5p + 25q = 450$ **[M1]**

Multiplying by $\frac{2}{5}$ gives $3p + 10q = 180$ **[A1]**

(ii) $p = 20$ **[B1]**

$q = 12$ **[B1]**

(c) **(i)** 24.25 **[B1]**

(ii) 24 **[B1]**

(iii) Upper quartile = 27 and lower quartile = 22 **[M1]**

5 **[A1]**

(d) Revised $\sum fm = 1245 - 12(25) + 291 = 1236$ **[M1]**

$\frac{1236}{100} = 12.36$ **[A1]**

[Back to Question] *[Total Marks: 12]*